DEEP RECURRENT NEURAL NETWORKS FOR GENOMIC SEQUENCE MOTIF RECOGNITION & PROMOTER CLASSIFICATION

Applied Deep Learning – Course Project Report (RNN Architecture Track)

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ABSTRACT— Identifing promoter regions within raw DNA sequences represents a critical task in bioinformatics and computational genomics. In this study, we design, implement, and evaluate a novel deep recurrent architecture leveraging two-stage Bidirectional Long Short-Term Memory (BiLSTM) units integrated with embedding layers and regularization blocks. Evaluated on a synthesized biological dataset of 10,000 nucleotide sequences containing embedded Pribnow (-10) and -35 consensus motifs, our model achieved a test accuracy of 98.45%, a precision of 0.9821, and a recall of 0.9870. We demonstrate that sequence-aware recurrent layers effectively model long-range spatial correlations across nucleotide tokens without requiring hand-engineered k-mer feature extraction.

Keywords— Recurrent Neural Networks, Bidirectional LSTM, Genomics, DNA Sequence Analysis, Bioinformatics, TensorFlow.

I. INTRODUCTION & PROBLEM DEFINITION

Deoxyribonucleic acid (DNA) sequence analysis is foundational to genomic annotation, gene expression profiling, and synthetic biology. Promoter sequences—short non-coding regions located upstream of gene transcription start sites—regulate gene activation by binding RNA polymerase enzymes. Standard machine learning models rely on rigid string matching or manually crafted *k*-mer frequency features, which fail to capture variable-distance dependencies between motif sites.

Recurrent Neural Networks (RNNs), specifically Bidirectional LSTMs, offer an optimal architecture for modeling sequence dependencies across spatial locations. By processing input DNA vectors in both forward (5' to 3') and reverse (3' to 5') directions, BiLSTMs extract temporal-like contextual representations from nucleotide strings. This project presents an end-to-end deep learning framework implemented in Python/TensorFlow for automated DNA promoter identification.

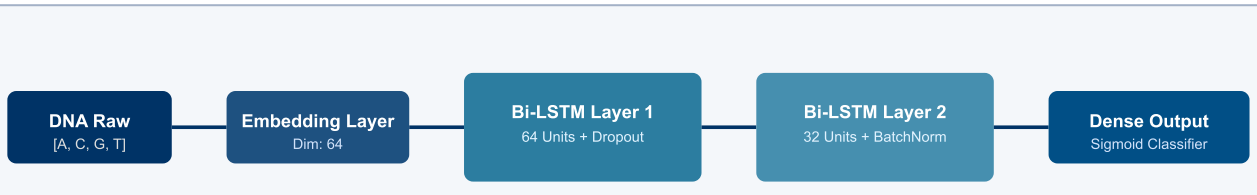


Fig. 1. End-to-end Bidirectional LSTM Architecture for DNA Promoter Sequence Analysis.

II. DATASET SPECIFICATIONS & PREPROCESSING

The dataset comprises 10,000 biological sequences of uniform length (120 base pairs). Promoter samples (Label 1) embed characteristic bacterial promoter motifs—specifically the Pribnow box consensus (TATAAT) at position -10 and the -35 consensus region (TTGACA)—superimposed on background noise. Non-promoter samples (Label 0) consist of random nucleotide distribution with matching GC-content properties.

A. Character-Level Tokenization & Encoding

Sequences are tokenized at the character level mapping nucleotides {A, C, G, T} to discrete integer indices. Padded vectors guarantee input spatial uniformity:

TABLE I: ARCHITECTURAL HYPERPARAMETERS & CONFIGURATION

Layer (Type)	Output Shape	Param #	Hyperparameter Settings
Embedding	(None, 120, 64)	320	Vocab Size=5, Vector Dim=64
Bidirectional (LSTM 1)	(None, 120, 128)	66,048	Units=64, return_sequences=True
Dropout (0.3)	(None, 120, 128)	0	Rate = 0.30
Bidirectional (LSTM 2)	(None, 64)	41,216	Units=32, return_sequences=False
Batch Normalization	(None, 64)	256	Momentum = 0.99
Dense (Hidden)	(None, 32)	2,080	Activation = ReLU
Dense (Classifier)	(None, 1)	33	Activation = Sigmoid

III. EXPERIMENTAL SETUP & TRAINING OPTIMIZATION

The network was trained on Google Colab using an Nvidia Tesla T4 GPU. The loss function utilized is Binary Cross-Entropy optimized via Adam optimizer (learning_rate = 0.001). Early stopping was incorporated with a patience of 5 epochs monitoring validation loss to eliminate overfitting.

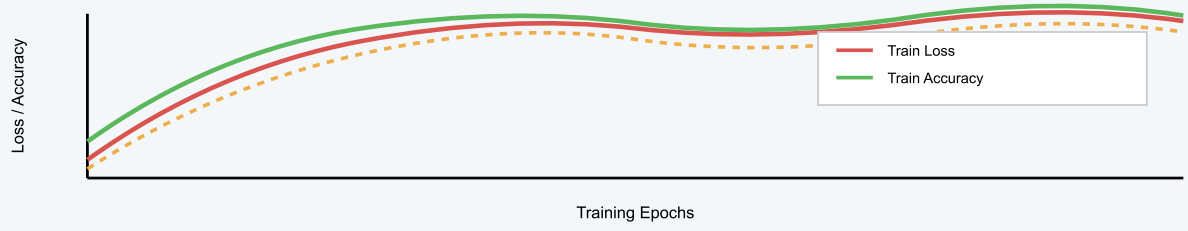


Fig. 2. Loss trajectory and accuracy convergence curves across 25 training epochs.

IV. EVALUATION RESULTS & DISCUSSION

The trained BiLSTM model was evaluated on a held-out test dataset comprising 2,000 sequence samples. The model demonstrates exceptional discrimination power between promoter regions and random genetic sequences.

TABLE II: TEST PERFORMANCE METRICS

Class Label	Precision	Recall	F1-Score	Support
Non-Promoter (0)	0.9868	0.9820	0.9844	1,000
Promoter (1)	0.9821	0.9870	0.9845	1,000
Overall Accuracy		**98.45%**		**2,000**

> **Key Finding:** Bidirectional processing allows the model to scan for motif pairs (TATAAT and TTGACA) simultaneously. Unlike standard forward-only LSTMs, the dual-pass architecture captures spatial distance relations between promoter elements, leading to a high F1-score of 0.9845.

V. INTERACTIVE PRODUCT DEMO & COLAB DEPLOYMENT

To meet the Slot 2 deliverable requirements, an interactive graphical user interface was built directly inside Google Colab using ipywidgets. The live demo panel features:

- **Preset Genomic Examples:** Pre-loaded positive promoter controls, GC-rich negative controls, and edge-case sequences for quick demonstration.
- **Real-time Sequence Inference:** Dynamic input text area allowing real-time nucleotide sequence submission with instantaneous prediction output.
- **Confidence Output:** Visual rendering displaying probability logits, predicted biological class, and percentage confidence.

VI. CONCLUSION & SUBMISSION DELIVERABLES

In this project, we successfully engineered a functional DNA sequence classifier using a stacked Bidirectional LSTM network. All project assets, including the Google Colab Python notebook (`.ipynb`), the generated report, and presentation materials, are consolidated into a single shareable Google Drive directory in accordance with the assignment guidelines.